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Novel PEGylated deep eutectic solvents: A thermophysical and photophysical investigation

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ABSTRACT

Deep eutectic solvents (DESs) have gained significant interest among researchers in the past two decades due to their improved physical and chemical characteristics beyond traditional organic solvents. They represent a recent advancement in eco-friendly solvents. Studying their physical characteristics would thus be very beneficial for designing methods in varied application domains. A majority of DESs are traditionally categorized into several types. Amongst them, the recently known *Type V* DESs have received increased attention. They are made using nonionic precursors that frequently exhibit marginal high vapor pressure and lower viscosities than their ionic equivalents and are chloride-free. In this study, four new PEGylated DESs are successfully prepared by using polyethylene glycol 200 (PEG200) as a hydrogen bond donor (HBD) with varied organic acids such as levulinic acid (LA), cinnamic acid (CIA), hydrocinnamic acid (HCA), and citric acid (CA) as hydrogen bond acceptor (HBA) in different molar ratios. Various atomic interactions between the HBAs and HBD in these prepared DESs were explained using Fourier transform infrared (FT-IR) spectroscopy and ^{1}H- and ^{13}C-nuclear magnetic resonance (^{1}H- and ^{13}C NMR) spectroscopy. Also, the temperature-induced transition in the prepared DESs was recorded using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). The thermophysical properties, such as density (ρ) and viscosity (η), were defined as a function of temperature. Also, the study investigates the physical and chemical properties of PEGylated DESs, focusing on their dipolarity/ polarizability, solvatochromic behavior, chemical structure, and composition. It uses Kamlet-Taft parameters to explore the H-bond donor and acceptor tendency and understand the role of acidic constituents. The study also investigates the solvation dynamics and diffusion of fluorescence quenching of pyrene and benzo [a]pyrene using nitromethane as a quencher. Here, the fluorescence lifetime adheres to the Stern–Volmer fitting, and higher bimolecular quenching rate constants support the faster diffusion of fluorophore and quencher molecules in these PEGylated DESs compared to other hydrophilic DESs. The study establishes constituent dependence and faster solvation dynamics.

1. Introduction

Deep eutectic solvents (DESs) are composed of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) having low lattice energy, held together due to the formation of H-bonding, resulting in a significant decrease in the melting point of the mixture compared to the individual components [1,2]. Along with the intricate H-bonding interaction, the ionic interactions, van der Waals forces, and the entropy of mixing also play a substantial role in the formation and depression in the melting point of DESs [2]. Over time, different kinds of DESs have been prepared using various compounds. Based on this, DESs are categorized into four primary groups according to their components: *Type I* includes quarternary ammonium salts and metal chloride. However, their utilization is limited by the high moisture sensitivity of these solvents. *Type II* consists of a quaternary ammonium salt and metal chloride hydrate. *Type III* DESs consist of a quaternary ammonium salt and

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organic molecule, such as amide, carboxylic acid, or polyhydroxy substituents. This class of DESs has acquired immense popularity because it is readily available in large libraries of constituents and their low toxicity. *Type IV* DESs comprise a metal chloride hydrate and an organic molecule. *Type V* DESs include nonionic molecules as hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD). In contrast to *Type III* DESs, *Type V* DESs are mostly non-polar due to the hydrophobic nature of the constituents consisting of long alkyl chains. Although the ionic contribution is absent, they exhibit a significant depression in their melting point due to prevalent H-bonding [3]. These DESs are often acknowledged as a class of ionic liquids (ILs) as they possess similar characteristics, which include high thermal stability, low volatility, low vapor pressures, and tunable polarity. Unlike ILs, inclined to be expensive, non-biodegradable, and highly toxic, DESs are typically safe, inexpensive, sustainable, biodegradable, non-toxic, and easy to prepare [3]. Such a wide range of observed properties enables DESs to replace the existing volatile organic compounds (VOCs) that are potentially used widely throughout the research and industrial sectors.

Given the immense number of combinations of HBA and HBDs, obtaining the optimal DES requires an exhaustive trial-and-error through many possible binary combinations. Such an outlook obstructs the adaptation of DESs for specific applications and effectively avoids many potential advantages that DESs offer. Though in the last two decades, DESs have gained significant research interest in biotechnology applications [4], flexible thermal sensors [5], CO_2 capture and conversion [6], surfactant self-assembly [7], formation of hydrophobic eutectogels [8], antimicrobial activity [9], liquid–liquid microextraction method [10], electrochemical process [11]. Despite numerous studies, there is still insufficient information covering the formation and characteristics of DESs.

DESs composed of nonionic compounds are simply considered to be ideal mixtures [12]. For instance, in the polyethylene glycol 200 (PEG200)-carboxylic acid system, a strong interaction arises due to the difference in acidity between the ether group and aliphatic hydroxyl and carboxylic groups of acids [13,14]. These DESs are considered inexpensive and offer eco-friendly preferences due to the presence of PEG components. They have been widely used as efficient absorbents for separating nitrogen and sulfur impurities from fuels [15] and recovering metals [16]. Additionally, PEGylated DESs are very effective solvents in material synthesis [17,18]. Amarpreet Singh et al. produced DESs by incorporating glycerol and PEG with choline chloride (ChCl) at different molar ratios. Their investigation has shown that the thermophysical characteristics of DESs may be tailored by altering the type of HBDs and controlling the molar ratios [13]. Filipa Lima et al. formulated various DESs utilizing PEG and its precursors and evaluated their efficacy in extracting sulfur compounds from a model fuel [15].

Before DESs and conventional solvents are compared to be employed industrially, it is necessary to quantify their solvation characteristics, such as H-bonding ability and polarity, along with the discernible physical parameters, for instance, melting point, boiling temperature, and density. The most comprehensive and widely exercised quantitative assessment of solvent photophysical characteristics is the Kamlet-Taft solvatochromic parameter set that measures the acidity (hydrogen bond donating ability), basicity (hydrogen bond accepting ability), and dipolarity/ polarizability. In this regard, quenching of fluorescence is one of the most reliable tools for getting insight into any newly developed solvent system. It offers a practical method to gain an understanding of accessibility dynamics and solute solvation behaviour within a solubilizing medium. In this direction, frequently used neutral fluorescence probes are pyrene and its derivatives [19,20].

In the current study, we attempt to prepare four DES systems based on PEG200 as an HBD and organic carboxylic acids, *i.e.*, levulinic acid (LA), cinnamic acid (CIA), hydrocinnamic acid (HCA), and citric acid (CA) as HBAs. Various thermophysical characteristics, such as viscosity, density, and other associated properties, were measured for PEGylated DESs. Spectroscopic techniques like FT-IR, ^{1}H- and ^{13}C NMR were employed to develop a comprehensive and clear understanding of the variations in solute–solvent interactions within the examined PEGylated DESs. To investigate the stability and thermal behavior of PEGylated DESs, thermogravimetric (TGA) and differential scanning calorimetry (DSC) analyses were performed. UV–Vis absorbance and fluorescent probes estimated the corresponding empirical parameters based on their solvatochromic responses. Also, we have investigated the fluorescence quenching of polycyclic aromatic hydrocarbons (PAHs) with nitromethane as a fluorescence quencher as a function of temperature. UV–Vis probes such as 4-nitroaniline (NA), *N*, *N*-diethyl-4-nitroaniline (DENA), and Nile Red (NR) were employed to calculate Kamlet-Taft parameters.

2. Experimental

2.1. Materials

Polyethylene glycol 200 (PEG200, 99.0 %) was acquired from SDFCL Pvt. Ltd., INDIA. Levulinic acid (LA, 98.0 %), cinnamic acid (CIA, 99.0 %), hydrocinnamic acid (HCA, 99.0 %), and citric acid (CA, 99.0 %) were acquired from Sigma-Aldrich, INDIA. Pyrene (99.0 %) and benzo [a]pyrene (99.0 %) were procured from Sigma-Aldrich, INDIA, and Accustandard, USA, respectively, and preserved in moisture-free conditions. Ethanol (99.9 %) and dichloromethane (99.0 %), used for stock solutions preparation of probes, were obtained from Merck, INDIA, and Sigma-Aldrich Co., INDIA, respectively. Nitromethane (99.0 %) was purchased from Acros Organics, INDIA. 4-nitroaniline (NA, 99.0 %) and *N*, *N*-diethyl-4-nitroaniline (DENA, 99.0 %) were purchased from Spectrochem Co. Ltd., INDIA and Frinton Laboratories, USA respectively, and Nile Red (NR, ≥98 % (HPLC)) was procured from Sigma-Aldrich, INDIA. The chemical structures of all ingredients used in this study are depicted in Table 1.

2.2. Preparation of DESs

Organic acids such as HBA and PEG200 as HBD were selected to prepare PEGylated DESs. The pure components were combined at a specific molar ratio (mentioned in Table 1), and the mixture was stirred at 333.15 K and 150 rpm until a clear and homogenous liquid was obtained. For weighing, Mettler-Toledo AB104-S balance (accuracy ±0.0001 g) was used. To prevent any alterations in the structure and physical characteristics of the DESs, the sample preparation underwent an environment controlled for moisture levels and stored in sealed containers. Fresh samples were used for testing to avoid any effects of humidity from the surrounding environment. However, before all experiments, we performed the Karl Fischer titration that observed the moisture absorption under 100 ppm. Here, the DESs (MDES) molar mass is calculated using Eq. (S1), and all other equations are given in Supporting information [21].

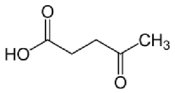
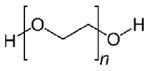
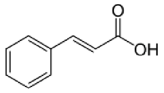
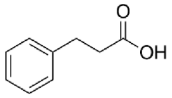
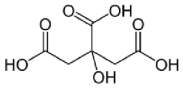
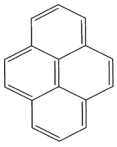
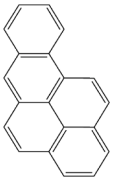
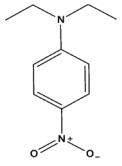
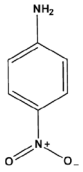
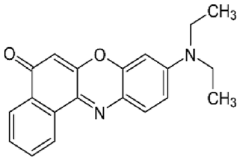
2.3. Spectral investigation

To determine the appropriate preparation of DESs, molecular structural analysis was conducted using a Fourier transform infrared spectrometer with the MIRacle 10, Single Reflection ATR accessory (Model: IR Affinity-1) over 500–4000 cm^{-1}. A 400 MHz JEOL instrument for ^{1}H NMR and a 100 MHz for ^{13}C NMR were employed at standard probe temperatures. The reference values were set at ~2.50 ppm for ^{1}H NMR and ~39.52 ppm for ^{13}C NMR using the residual internal DMSO-d_6.

2.4. Thermal analysis

The DESs were analyzed using an SDT Q600 thermal analyzer from TA Instruments, USA. The DESs samples, weighing approximately 2.0–2.5 mg, were placed in an alumina pan. They were then heated from 293.15 to 723.15 K in a nitrogen environment with a 99.0 mL/min flow

Table 1
Constitutional details of the ingredients proposed in this study.

Ingredients		Abbreviation	Molar ratios	M_{DES} (g/mol)
HBA	HBD			
Levulinic acid (LA)	Polyethylene glycol 200 (PEG200)	LA: PEG	1: 1	316.11
				
Cinnamic acid (CIA)		CIA: PEG	1: 4	948.16
				
Hydrocinnamic acid (HCA)		HCA: PEG	1: 1	350.17
				
Citric acid (CA)		CA: PEG	1: 2	592.12
				
Polycyclic aromatic hydrocarbons (PAHs)		**UV–Vis probes**		
				
Pyrene	Benzo[a]pyrene	*N*, *N*-diethyl-4-nitroaniline (DENA)	4-Nitroaniline (NA)	Nile red (NR)

rate. All the experiments were done at heating rates of 10 K/min. Data was processed and analyzed using the TA Universal Analysis 2000 software [22].

The differential scanning calorimetry (DSC) analysis was conducted using a Mettler Toledo DSC1/700 W equipped with an HSS8 sensor, a Huber TC100MT Intercooler, and an Olympus microscope attachment. The sample was placed in an aluminum pan with a pin as the reference. Experiments were conducted with a heating rate of 5 K min^{-1} under nitrogen flow [23].

2.5. Solution flow study

The density (ρ) and viscosity (η) of the prepared DESs were determined simultaneously using a digital density meter by Anton Paar (model: SVM 3001 viscometer). To ensure accuracy, this instrument was cleaned with ethanol and then toluene before and after each measurement session. The meter boasts a ρ repeatability of ± 0.00005 g m^{-3}. Given the high sensitivity of ρ to temperature changes, the integrated Peltier device proved crucial for temperature control during the measurements. Here, the temperature was maintained with an accuracy of ± 0.1 K.

2.6. Photophysics study

The stock solution of probes was prepared by the dissolution of the required quantity of each probe in ethanol and dichloromethane, and they were stored in clean amber glass vials at 277.15 K (4 °C). The required volume of stock solution was taken in a quartz cuvette with a 1 cm path length, and the solvent was evaporated using highly pure grade 1 nitrogen. The pre-requisite DES amount was transferred to the cuvette. UV–Vis absorbance spectra of DENA, NA, and NR were collected using a Lambda 35 double beam spectrometer (procured from PerkinElmer) that has a modifiable bandwidth and contains deuterium and xenon lamps and a photomultiplier tube (PMT) detector.

The steady-state fluorescence spectra were acquired on the instrument (purchased from Edinburg Instruments Ltd.) with an FLS1000 spectrometer that is outfitted with STGM325-M emission monochromators and STGM325-X grating excitation and contains a 450 W xenon arc lamp as a source of radiation, a red PMT detector and a quantum northwest luma 40 to control temperature. Before data analysis in Sigma Plot v14 software, all the spectra were properly corrected by subtracting the optical responses from appropriate blanks. Inner filter correction in fluorescence intensities was done according to Eq. (S11).

Time-resolved excited-state intensity decay measurements of pyrene were acquired using Horiba-Jobin Yvon, Inc., fluoro cube time-correlated single photon counting (TCSPC) fluorimeter. 340 nm UV-pulsed nano LED-340 source having a pulse width <1.0 ns was used to excite the probe. DAQ-MCA-3 series (P7882) multichannel analyser was exercised to collect data. The software subtracted the instrument response function (IRF) of a scattering solution of glycogen in water. The excited-state intensity decays were analyzed using DAS6 analysis software and were fitted to desired decay models.

3. Results and discussion

3.1. Spectral analysis

Inference of the functional groups and characteristic absorption profile for the prepared DESs using FT-IR in the range of 4000–500 cm^{-1} are depicted in Fig. 1. Distinct absorption bands, as depicted in Fig. 1a,

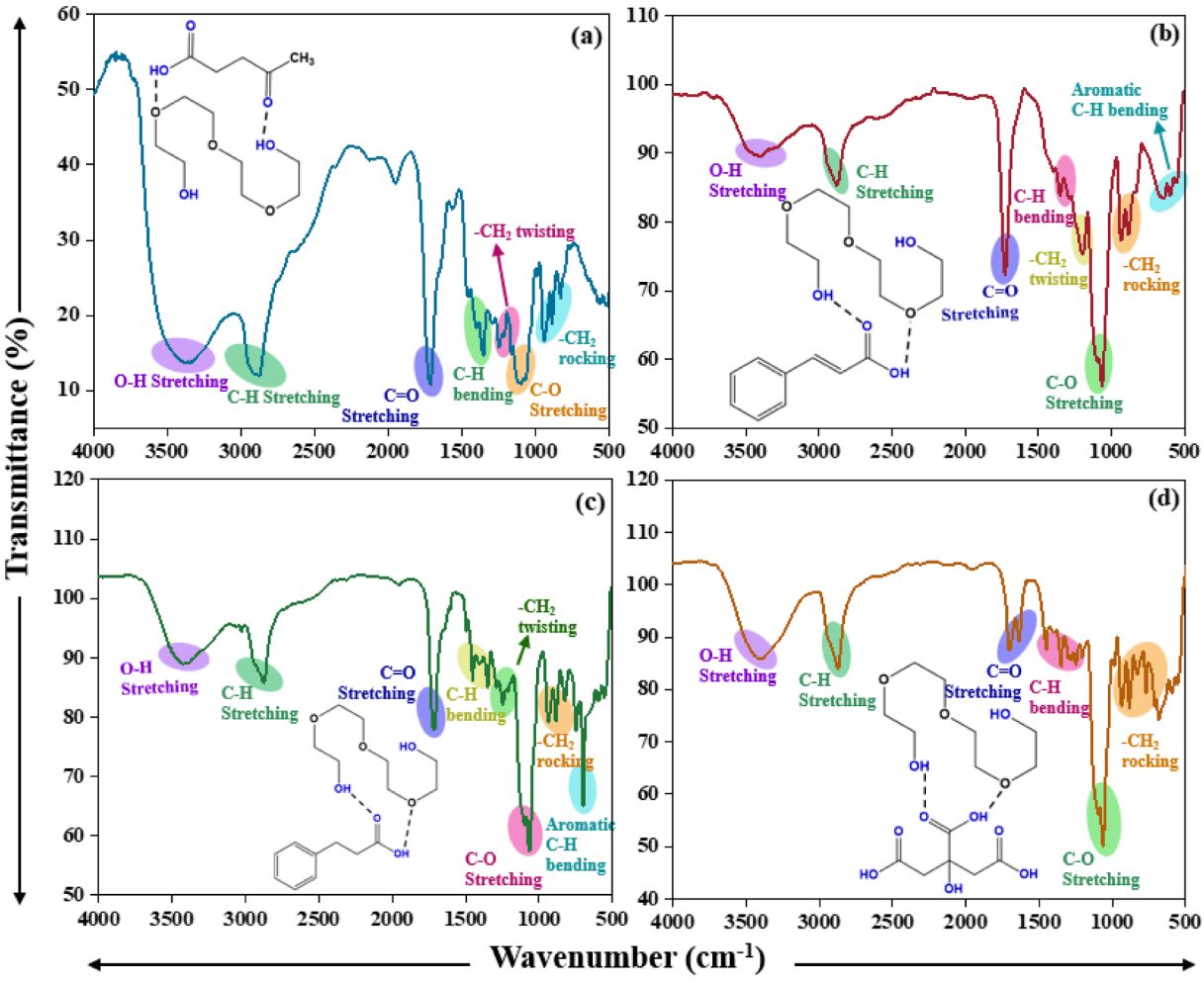


Fig. 1. Spectral profile for the prepared PEGylated DESs: (a) LA: PEG; (b) CIA: PEG; (c) HCA: PEG; and (d) CA: PEG at 298.15 K.

have been recognized based on the unique characteristics of LA: PEG, which includes the stretching vibration peak of O—H bond in PEG200 molecules at 3365 cm^{-1}, which, owing to the formation of H-bond in DES, aliphatic C—H stretching at 2870 cm^{-1}, strong C═O stretching at 1718 cm^{-1}, C—H bending vibrations, —CH_2 wagging at 1356 cm^{-1}, —CH_2 twisting at 1245 cm^{-1}, C—O vibration at 1103 cm^{-1}, —CH_2 rocking at 938 cm^{-1}, C—C stretching at 887 and 827 cm^{-1} [24].

In the case of CIA: PEG, as shown in Fig. 1b, FT-IR spectra peaked at 3403 cm^{-1}, indicating that O—H stretching of the alcohol group of the CIA and the H-bonding interactions between CIA and PEG. Peaks around 2875 and 1351 cm^{-1} confirmed the C—H stretching of the alkane group, while strong C═O stretching of conjugated acid regions showed a significant peak at 1705 cm^{-1}, C—H bending vibrations, and —CH_2 scissoring at 1452 cm^{-1}, —CH_2 twisting and the C—O vibration at 1250—1066 cm^{-1} (range), —CH_2 rocking, and an aromatic C—H out-of-plane bending vibration modes of C—C groups at 940–680 cm^{-1} are shown respectively.

The FT-IR spectra of HCA: PEG (Fig. 1c) presented the characteristic peaks of the stretching vibrations of —OH of the carboxylic group at 3417 cm^{-1} and indicated the H-bonding interactions between HCA and PEG, the C—H stretching of alkanes group at 2880 cm^{-1}, strong C═O stretching carboxylic acid at 1710 cm^{-1}, C—H bending vibrations, and —CH_2 scissoring at 1455 cm^{-1}, —CH_2 twisting at 1250 cm^{-1}, C—O stretching at 1200–1065 cm^{-1} (range), and —CH_2 rocking, and an aromatic C—H out-of-plane bending vibration modes of C—C groups at 940–825 cm^{-1} are observed. Likewise, the FT-IR spectra of CA: PEG shown in Fig. 1d depicted the characteristic peaks of the stretching vibrations at 3400 cm^{-1}, representing the significant presence of the formation of H-bonding in the DES as well as the —OH group contributed by the structure of CA which has three —OH groups attached in a single molecule. This correlates effectively with the data that has been documented. 2880 cm^{-1} represented sp^2 and sp^3 hybrid C—H bond stretching vibrations, strong C═O stretching carboxylic acid at 1728 cm^{-1}, C—O stretching at 1200–1066 cm^{-1}, —CH_2 rocking, —CH_2 twisting at 933–883 cm^{-1} are observed. Thus, the analysis of the prepared DESs revealed no variations in the position of bands or the emergence of new bonds in comparison to the pure acids, thereby indicating no significant chemical interactions between the components of DESs.

The proton and carbon nuclear magnetic resonance (^{1}H- and ^{13}C NMR) spectra, shown in Fig. 2(i–ii), provided further evidence of interaction between PEG200 and HBAs. Zhao et al., using ^{1}H NMR, revealed that the terminal hydroxyl groups on PEG form H-bonds, leading to eutectic formation [25]. Wenjun Chen et al. demonstrated that when HBA-like urea is added, the —OH hydrogen atoms in PEG shift downfield. This movement indicates the formation of H-bonds between PEG and HBAs. The hydrogen of the —OH group in thiourea shows a smaller downfield shift than other PEGylated DESs, suggesting a relatively weaker H-bonding interaction. Interestingly, the behavior of hydrogen atoms of the —OH group in acid-*based* DESs is different. In oxalic acid in PEGylated DES, the chemical shift moves upfield, while in other carboxylic acids, it vanishes, potentially due to proton transfer. This aligns well with the data that has been reported [26].

The ^{1}H NMR spectra for all the tested DESs are depicted in Fig. 2i. In

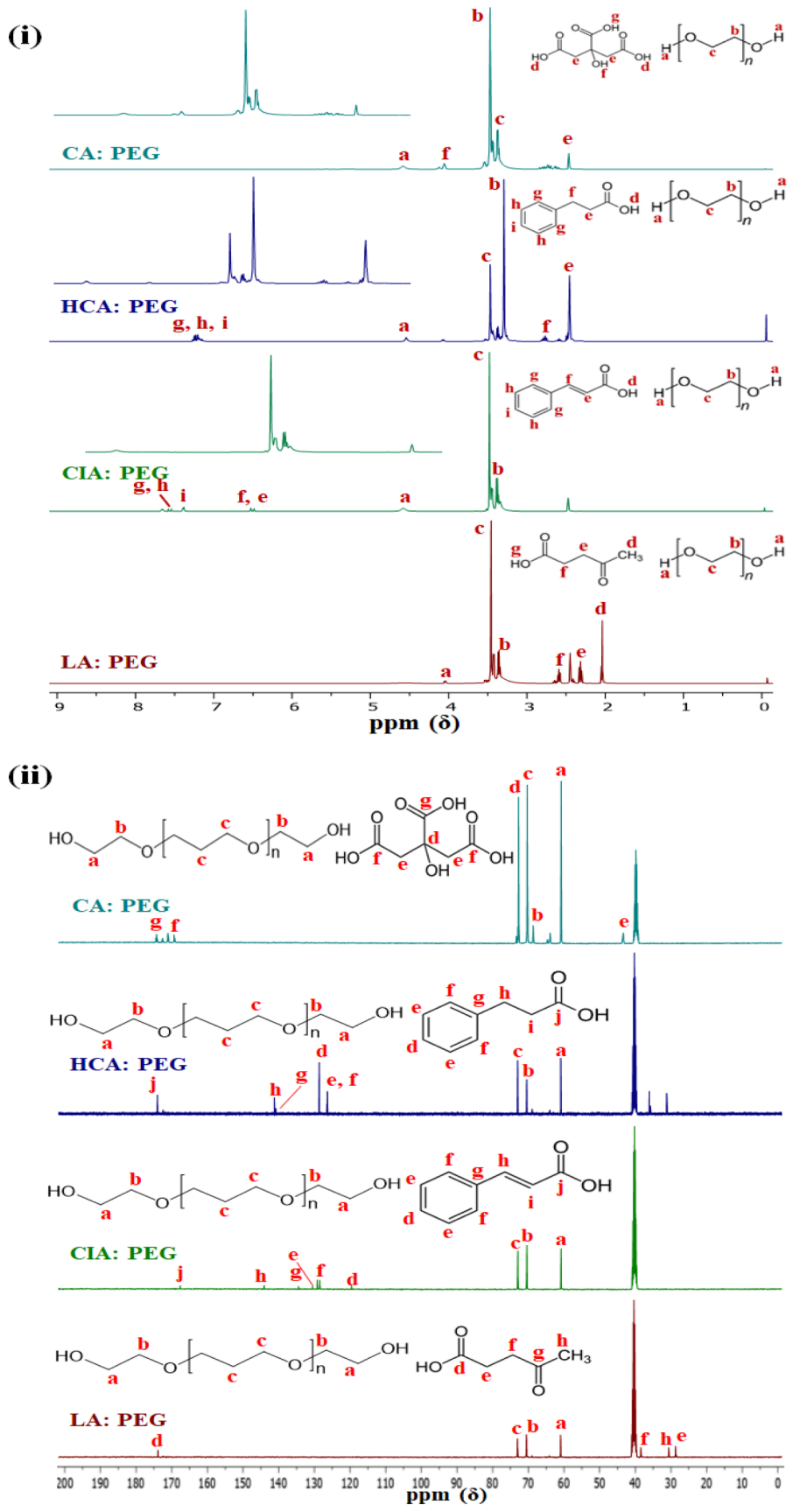


Fig. 2. (i) 1H NMR; and (ii) 13C NMR spectra of (a) LA: PEG; (b) CIA: PEG; (c) HCA: PEG; and (d) CA: PEG.

its pure form, the —OH group of PEG200 displayed a broad singlet peak at ~4.57 ppm, attributed to the consistent environment of its protons. In Fig. 2i, a distinct and intense hydrogen signal at approximately ~3.74 ppm is observed, linked to the methylene group adjacent to the hydroxyl group of PEG200. This signal shifts to ~3.5 ppm after DES formation. Carboxylic acids primarily exhibit a peak within the ~11–12 ppm range, although this peak might be broadened due to H-bonding and rapid solvent exchange, potentially rendering it less discernible. Thus, the observed chemical shifts indicated the formation of the H-bond between the carboxylic acids and PEG200 during DES formation, corroborating the results from FT-IR analysis. Also, the hydrogen atom associated with the aromatic rings of CIA and HCA shows its signal at ~7.53 ppm in its pure form. Specifically, it is located in a lower field at ~7.22 ppm in the case of CIA: PEG, while it moves to a higher field at ~7.60 ppm in the case of HCA: PEG.

As illustrated in Fig. 2i, there is a notable peak within the ~6–7 ppm range, which indicates the vinylic protons located next to the carbonyl group present in all the carboxylic acids. Furthermore, peaks near the ~2–3 ppm range can be attributed to the methylene (—CH_2) groups of carboxylic acids. Notably, the methylene group neighboring the carbonyl functional group tends to resonate slightly downfield, closer to ~3 ppm. The terminal methyl (—CH_3) group is characterized by a peak around ~2 ppm region.

A general breakdown of the typical ^{13}C NMR peak values for all prepared DESs is mentioned in Fig. 2ii. Here, the methylene carbon, which is linked by ether bonds, represented as —OCH_2CH_2- units, resonates in the ~60–70 ppm region. This is attributed to the carbon spectrum of pure PEG200. Also, the carbonyl carbon gives a distinct resonance between ~205 and 215 ppm, highlighting its deshielded state. The vinylic carbons resonate between ~120–140 ppm. The methylene and terminal methyl carbons of carboxylic acids show signals in the ~15–35 ppm range, revealing their saturated nature. The carbon bearing the hydroxyl group can be found in the ~70–80 ppm region, showcasing the effect of the electron-rich oxygen atom. As shown in Fig. 2ii, aromatic carbons exhibit resonances in the range of ~120–140 ppm, influenced by the conjugated double bond and the electronic environment of the aromatic ring observed for CIA and HCA. For HCA: PEG, adjacent to the aromatic ring, the methylene carbon, known as the β-carbon, shows up in the ~25–35 ppm range.

3.2. Thermal analysis

Studies have reported the thermal stability evaluation of DESs to be one of the most crucial subjects of research. For instance, Ayoub et al. utilized TGA/ DTG techniques and deduced that the thermal stability of DESs is more significantly impacted by HBD than HBA. They observed that a longer alkyl chain in HBD enhances the thermal stability of DES. Moreover, they determined that as the proportion of HBD in DES increases, its thermal stability tends to decrease [27]. Furthermore, on PEGylated DES, Yu Chen et al. revealed significant thermal stability at the moderate temperature of 353.15 K over 24 h period [16].

Fig. 3 represents the weight loss measured in % against temperature for the studied DESs. Fig. 3a shows a weight loss of 98 % observed between 327.15 and 481.15 K for LA: PEG. In CIA: PEG, a weight loss of 96 % is observed between the temperature range of 388.15–551.15 K (Fig. 3b). As can be seen in Fig. 3c for HCA: PEG, the curves consist of three steps: the first step shows the decomposition from 333.15 K to 408.15 K could reach ca. ~23 % mainly attributed to evaporation of PEG200, while the second step corresponds from 408.15 K to 508.15 K may lose up to ~61 % mainly because of the evaporation of HCA, and finally the third step for least ~10 % from 508.15 K to 596.15 K which may be for the melting point of HCA: PEG. Interestingly, a significant

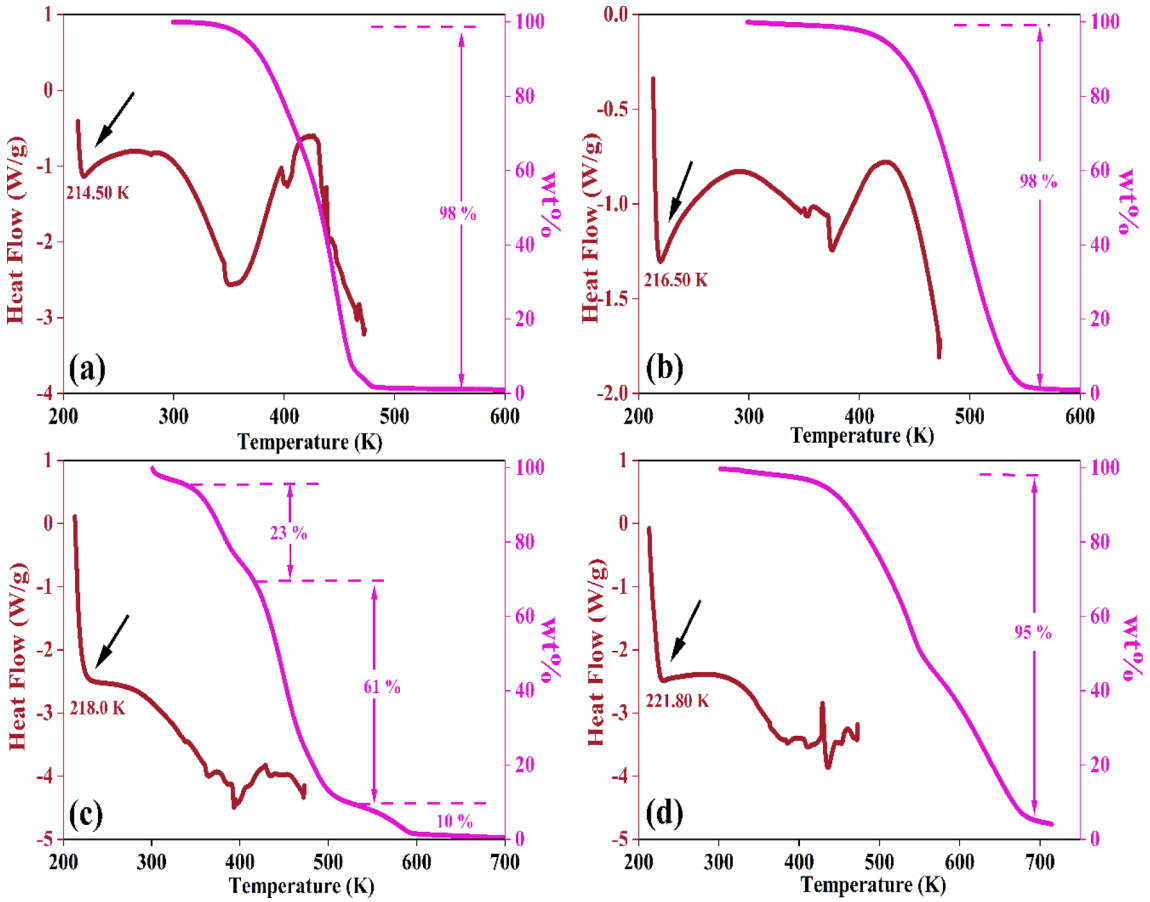


Fig. 3. DSC and TGA thermograms of PEGylated DESs (a) LA: PEG; (b) CIA: PEG; (c) HCA: PEG; and (d) CA: PEG.

weight loss of 93 % was determined for the DES CA: PEG between 404.15 and 697.15 K (Fig. 3d), which corresponds well with the reported data [17,28–30]. Thus, the thermograms of the prepared PEGylated DESs typically exhibit a singular phase of weight reduction, except for HCA: PEG, which displayed additional plateaus.

Many in the DES community are now debating the accuracy of the term "deep eutectic". However, for instance, Wenjun Chen et al. studied carboxylic acid and amide-*based* PEGylated DESs using DSC analysis, and their results revealed that the melting points of all PEGylated DESs were lower than those of their pure components, which they attributed to the formation of hydrogen bonds between the different components. In our study, we too performed DSC experiments and observed that the melting point of the tested HBD, *i.e.,* PEG200, is 223.15 K, and for four HBAs, *i.e.,* LA, CIA, HCA, and CA, is 306.15 K, 406.15 K, 320.15 K, and 426.15 K, respectively (Fig. 3a–d). The DSC thermograms as illustrated showed the melting points of all PEGylated DESs to be lower than those of their individual components, justifying the successful preparation of PEGylated DESs [18,26].

Thermophysical properties

Understanding the physical characteristics of solvents, such as their thermal stability and density (ρ) measurements, plays a crucial role in thermodynamic computations and process equipment sizing. Table S1, in the Supporting information, presents the ρ values of PEGylated DESs solutions prepared, which were measured at ambient pressure and within the temperature range of 283.15–363.15 K. Table S2 presents the fitting parameters (A and B) of linear equations derived empirically from least-squares analysis, which illustrate the impact of temperature on ρ. The excellent correlation between these properties and temperature is evidenced by the high values of the correlation coefficient (R^2). Here, the temperature dependence was modeled using a linear Eq. (S2). Fig. 4 illustrates the ρ of the DES that showed a linear decrease as a function of temperature. The fitting parameters (A and B) were used to describe this relationship, where A and B are unique for each DES [31].

In this work, the second-order polynomial fitting was correlated to define the experimental ρ of DESs as a function of temperature by applying Eq. (S3) as reported from the literature [31–33]. After analyzing the impact of various organic acids on DESs, it was stated that

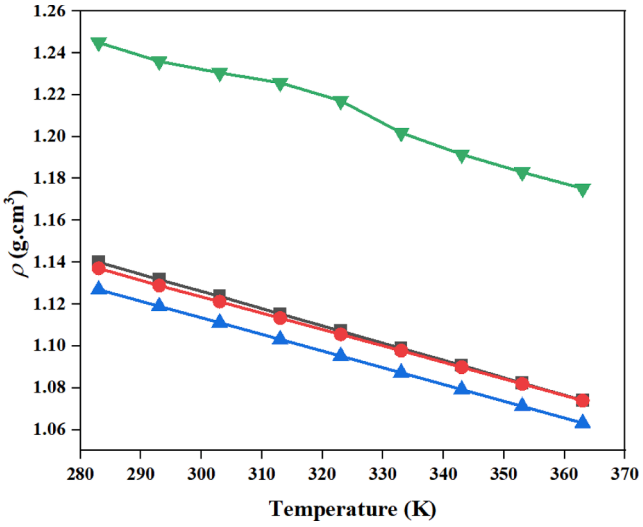


Fig. 4. Temperature behavior on ρ of various DESs: (a) LA: PEG (black); (b) CIA: PEG (red); (c) HCA: PEG (blue); and (d) CA: PEG (green). (The points in the figure represent the experimental values while the solid lines correspond to the calculated values correlated from Eq. (S4)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

DESs containing CA have a higher value of ρ than those containing other organic acids. This is likely since CA has three carboxylic acid groups, while the other three organic acids only have one. Interestingly, DESs prepared from LA and CIA, with one carboxylic group, have identical ρ at the same temperature. This approach assumes the isobaric thermal expansion coefficient (α_p) remains constant at a specific pressure. Table 2 presents the α_p values using Eq. (S4) for all the prepared PEGylated DESs.

The empirical linear equation for a given DESs was obtained by creating a straight line by plotting ln ρ against T using Eq. (S5). The linear fitting of this plot exhibits a coefficient of regression exceeding 0.99 and standard deviations within the experimental error (represented in Fig. S1 of Supporting information).

The calculation of molecular volume (V_m) for DESs solutions in this study involves adding the volumes of the HBA and HBD. The Eq. (S6) is used to determine V_m. The molar volumes were determined by analyzing the ρ results within a temperature range of 283.15–363.15 K, and the data is summarized in Table 2 [32]. It is a predictable trend that the V_m of each compound increases proportionally with its molar mass. The standard entropy (S°) [34], and lattice potential energy (U_{POT}) [35] are calculated from Eqs. (S7) and (S8), respectively. High lattice potential energy signifies thermodynamic stability; thus, LA: PEG appears to be the most stable DES among the prepared PEGylated DESs.

The significance of viscosity (η) lies in its impact on the transport phenomena of fluids [36]. Also, η profoundly affects the movement of dissolved particles within the solvent. Therefore, a better understanding of η is a crucial approach in determining the suitability of fluid for various applications. However, their high η is a significant disadvantage. The primary reason for the high η of DESs is their extensive H-bond network that restricts the mobility of free species within the solvent. This information is widely accepted in the scientific community [32,37].

The experimental η data for all PEGylated DESs solutions were observed at various temperatures and are illustrated in Fig. 5 and Table S1. Based on our experimental results, it can be inferred that the CA: PEG has a much greater viscosity value when compared to the remaining three DESs. When the temperature rises, the η tends to decrease significantly. An Arrhenius model was utilized to fit the η data. Eq. (S9) shows the relationship between η and T for the four different DESs. Fig. S2 shows the curves of log η plotted against 1/T for each of the four prepared DESs [32].

The measurements of η for PEGylated DES solutions were conducted over a range of temperature from 283.15 to 363.15 K. To study the temperature-dependent η behavior of these DES, the Vogel–Fulcher–Tammann (VFT) model was used. The VFT model is represented by Eq. (S10). The η is measured in mPa.s and is influenced by the adjustable parameters A_η, B_η, and C_η, which are obtained through experimental fitting and presented in Table 3. According to the results, an increase in temperature resulted in a decrease in η. This can be explained by decreased H-bonding interactions between the DESs complexes as the temperature increases. This result is consistent with the reported work [32]. Based on the values of R^2, it was concluded that the VFT equation represents a better fitting for the dynamic viscosity results.

Table 2
Summarization of magnitude of thermophysical, structural, and chemical properties.

DES	$\alpha_p \times 10^{-4}$	V_m (nm^3)	S°	U_{POT}
LA: PEG	−7.4499	0.46	602.89	406.88
CIA: PEG	−7.1154	1.39	1762.14	313.78
HCA: PEG	−7.2759	0.65	839.73	395.62
CA: PEG	−7.4313	0.80	1026.70	357.00

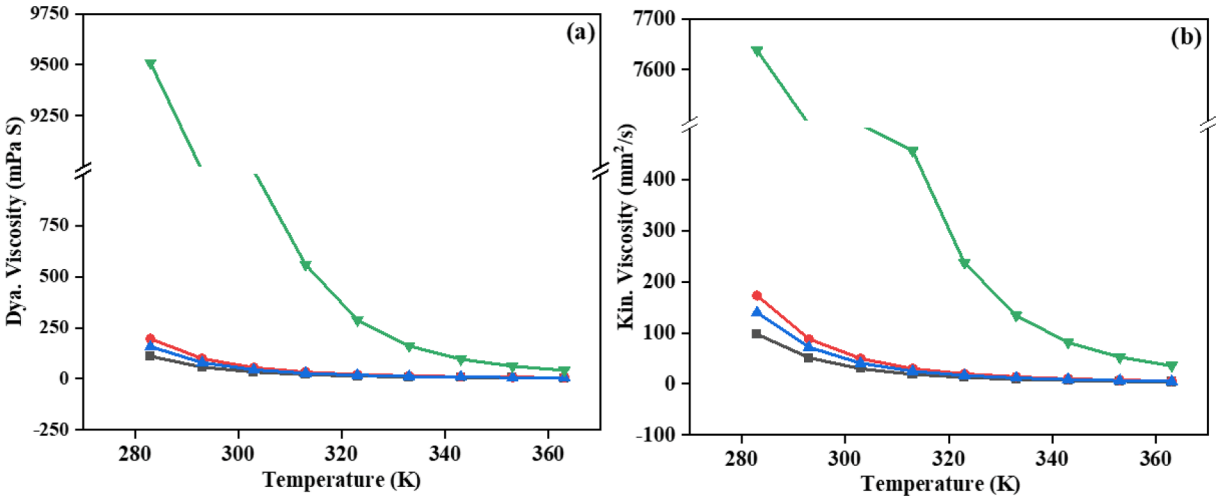


Fig. 5. Plot of (a) Dynamic viscosity; and (b) Kinetic viscosity versus temperature for the PEGylated DESs: (i) LA: PEG (black); (ii) CIA: PEG (red); (iii) HCA: PEG (blue); and (iv) CA: PEG (green) as a function of temperature. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Fitting parameters evaluated using several equations and their respective standard error and regression coefficient.

DES	LA: PEG	CIA: PEG	HCA: PEG	CA: PEG
Arrhenius equation				
E_a	−34.1523	−83.9570	−81.8417	−132.2172
η_0	−4.3300	−10.3798	−10.2168	−15.5278
R^2	0.9912	0.9920	0.9911	0.9904
Vogel–Tamman–Fulcher (VTF) equation				
A_η	−1.1353	−1.1811	−1.1178	−1.2981
B_η	4.3117	4.7682	4.2733	6.5179
C_η	171.9686	177.4317	181.5442	184.3148
R^2	0.9999	0.9999	0.9999	0.9999
E_a	200.1	241.8	232.2	371.7

3.3. Photophysics study

For the fundamental characterization of solvatochromic properties of new solvents, the estimation of Kamlet-Taft solvatochromic parameters (α, β, and π*) is one of the primary steps [38]. These prevalently exercised parameters reveal the physicochemical behavior of the solvent toward the solute [39,40]. It also represents the H-bond donating ability (acidity), H-bond accepting ability (basicity), and dipolarity/polarizability of solvent, respectively. In our study, NA, DENA, and NR are commonly employed to calculate these solvatochromic parameters.

The UV–Vis absorbance spectra of these probes dissolved in the investigated DESs except CIA: PEG because of their yellow coloration and high absorbance value. As shown in Fig. 6, the respective absorbance maxima (λ_{max}^{abs}) values are listed in Table 4. We have estimated the values of α, β, and π* for DESs using the absorbance spectral features and Eqs. (S12)–(S14). The π* parameter indicates the ability to exhibit non-specific interactions with the solute, *i.e.*, dipole–dipole and dipole-induced dipole interactions. All three DESs of interest exhibit high polarity (π*) values. The carboxylic acid (—COOH) functional group contributes to the high polarity values of these DESs. The π* value for LA: PEG and CA: PEG is almost equal (0.99), indicating similar dipolarity for

Table 4
Absorption maxima λ_{max}^{abs} (nm) of DENA, NA, and NR and Kamlet–Taft polarity parameters of prepared DESs.

DES	LA: PEG	HCA: PEG	CA: PEG
Absorption maxima			
DENA	410	404	410
NA	375	373	381
Nile Red	564	556	566
Kamlet-Taft polarity parameters			
α	0.80	0.77	0.86
β	0.43	0.52	0.58
π*	0.99	0.88	0.99

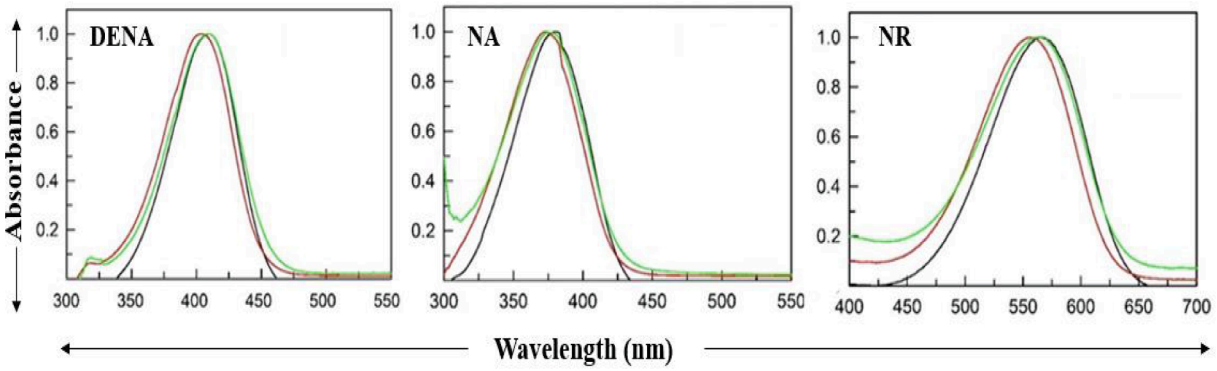


Fig. 6. Variation in UV–Vis absorbance for the probes (DENA, NA, and NR) dissolved in DESs: (a) LA: PEG (green); (b) HCA: PEG (red); and (c) CA: PEG (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

these two DESs. Although our DESs are formed using neutral constituents. The π* values are comparable to many DESs reported in the literature [41]. Also, as compared to the other recurrently used *Type III* and *Type V* DESs. Harifi et al. conducted an experimental investigation into the solvatochromic properties and ion solvation structure of *Type III* DESs using PEG 400 as a co-solvent. Their results, supported by molecular dynamics (MD) simulations, revealed that PEG 400 forms strong H-bonds with these DESs, facilitating their incorporation into the solvation sphere of solvatochromic probes. Additionally, both experimental and theoretical studies were carried out to explore the preferential solvation of probes in aqueous solutions of *Type III* DESs and their individual components [42,43]. The ability of the solvent to donate —H in solvent-to-solute H-bonding is denoted by α parameter. The highest α value of ~0.86 was observed for CA: PEG and it is ascribed to the presence of one —OH and three —COOH moieties in CA. The H-bond acidity of all three DESs is found to be higher than ionic liquids [44], and alcohols and slightly lower than water [44,45]. The capacity of the solvent to accept H-bond is denoted by the β parameter. Here, β value for CA: PEG is higher (0.58) than HCA: PEG (0.52) and LA: PEG (0.43); this is due to the presence of three carboxylic acid moieties that can take part in extensive H-bonding. The polar carboxyl groups in acids contribute to the higher polarity of acid-*based* DESs than amide- and ammonium-*based* DESs. Chen et al. have exploited NR to determine the polarity of several PEGylated DESs [26]. It was reported that λ_{max}^{abs} values of NR in the amide- and ammonium-*based* DESs lie within the 555–558 nm range, implying the high polarity for the acid-*based* DESs. These results are in coherence with our outcomes, suggesting that the presence of —COOH moiety increases the polarity of a DES [46,47].

The fluorescence-*based* techniques to understand solvation in intricate solubilizing media have gained significant attention because of their intrinsic sensitivity. We have used two distinct alternate PAHs (Pyrene and Benzo [a]pyrene) with multidimensional responses to examine potential interactions, such as solute–solvent, solute–solute, and solvent–solvent in the perspective of solvent dipolarity within the PEGylated DESs. We intend to investigate probing molecular environment-dependent interactions inside our DESs using this methodology so that the results can explain the behaviour of solutes with related functions.

Based on the pyrene I_1/I_3 values, it was inferred that the highest polarity is observed for CA: PEG (1.89), trailed by LA: PEG (1.80) and HCA: PEG (1.66). The observed trend of DES polarity using pyrene further corroborates with the dipolarity/polarizability measurements performed using NR dye. The I_1/I_3 values for these DESs are considerably lower than *Type III* DESs (reline, 2.51, and maline, 2.18) [48]. The 1.70 pyrene I_1/I_3 value for PEG200 indicates the significance of the counter component of a DES; both CA and LA contribute towards the increased polarity of the DESs, whereas in the presence of HCA, decreased polarity is observed. It should be noticed that the molar ratio of LA: PEG and HCA: PEG is (1:1) while that of CA: PEG is (1: 2). Consequently, an explicit comparison with CA: PEG is not possible; however, the other two DESs share similar composition ratio and can be compared to access their effects on the physicochemical properties of their respective DESs.

Steady-state fluorescence quenching of pyrene and benzo [a]pyrene by nitromethane is demonstrated in Fig. 7. In the case of pyrene without quencher, emission peaks only correspond to monomeric species, *i.e.*, these DESs do not support excimer formation. Also, in the presence of a quencher, there is no evidence of fluorescent exciplex formation (Panel A–C), indicating that no significant solute–solute interactions are taking place, and these probes can be essentially utilized for studying solute–solvent interactions. Among the two investigated probes, the higher quenching efficiency was observed for pyrene, which prompted us to do a thorough temperature-dependent investigation of pyrene quenching within these DESs [49,50].

As established in the literature, pyrene excited-state is quite sensitive to exogenous additives in the solubilizing milieu because of its prolonged excited-state lifetime. Various quenching agents (molecular, atomic, and ionic) for pyrene fluorescence by different quenching methods have also been well-documented [19,20]. The one among them that is most frequently used to study the dynamics and solvation characteristics of the solvent systems is nitromethane. When pyrene is excited, its lowest unoccupied molecular orbital (LUMO), which becomes occupied, has a higher energy than the LUMO of the nitromethane. As a result, nitromethane serves as the electron/charge acceptor and excited-state pyrene as an electron/charge donor during the quenching process [20,51]. To understand the dynamics and mechanism of pyrene quenching within these PEGylated DESs, excited-state emission intensity decay [I(t)] of pyrene dissolved in the investigated DESs was studied using time-resolved fluorescence measurements with or without a quencher. The data was collected with various quencher concentrations as a function of temperature. It was observed that the I(t) of pyrene showed a best fit to a double-exponential decay

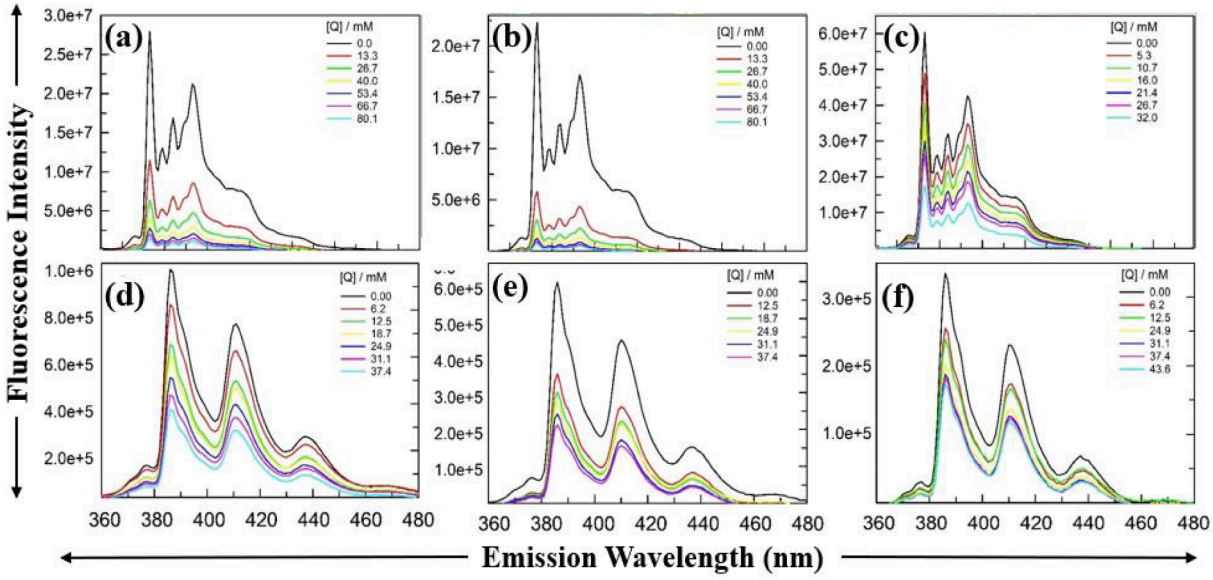


Fig. 7. Fluorescence emission intensities of pyrene (excited at λex = 337 nm) dissolve in (a) LA: PEG; (b) HCA: PEG; and (c) CA: PEG and Benzo[a]pyrene (excited at λex = 350 nm) dissolved in (d) LA: PEG; (e) HCA: PEG; and (f) CA: PEG with varying concentration of nitromethane as a quencher.

expression, as Eq. (S15), in the presence of the quencher regardless of temperature (298.15 and 338.15 K) and chemical constituents of DESs.

The double exponential decay of pyrene I(t) upon adding nitromethane corroborates with the literature as well [51]. The representative plots of pyrene emission intensity decay at 298.15 K and 328.15 K in the absence and presence of nitromethane (32 mM) are presented in Fig. 8a–c. The lifetime decay plots reveal the effect of temperature on the lifetime of pyrene in all three investigated DESs. Out of the two recovered decay lifetimes of pyrene at a fixed temperature, the shorter decay time does not vary systematically with nitromethane concentration. In contrast, the longer lifetime (τ) decreased monotonically with the increased quencher concentration. It is well-established that the longer excited-state lifetime allows the mutual approach of M* (excited pyrene molecule) and Q (quencher molecule). The bimolecular excited-state event is thus controlled by diffusion, and this type of quenching is called dynamic quenching [19].

At each of the investigated temperatures for every DES investigated (Fig. 9), the linear plots of τ_0/τ versus [Q] (quencher concentration) demonstrate the dynamic nature of fluorescence quenching resulting in equivalent Stern–Volmer quenching constant (K_{SV}) and dynamic quenching constant (K_D) calucated using equations S16 and S17. The estimated K_{SV} and bimolecular quenching rate constant (k_q) for the pyrene-nitromethane fluorophore-quencher pair in the investigated DESs in the temperature range 298.15–328.15 K are listed in Table 5.

The k_q for the pyrene-nitromethane pair varies between 0.1×10^9 M^{-1} S^{-1} (for CA: PEG at 298.15 K) to 3.4×10^9 $M^{-1}S^{-1}$ (for HCA: PEG at 338.15 K). The k_q values are also compared to other relevant DESs with similar viscosities [52]. In our investigated DESs, dynamic η varies from 57.74 mPa.s at 293.15 K to 9.90 mPa.s at 333.15 K for LA: PEG, 80.03 mPa.s at 293.15 K to 12.72 mPa.s at 333.15 K for HCA: PEG DES and it ranges from 3161.00 mPa.s at 293.15 K to 160.91 mPa.s at 333.15 K for CA: PEG DES respectively. At 298.15 K, the k_q values vary in the following order: LA: PEG (1.0×10^9 $M^{-1}S^{-1}$) > HCA: PEG (0.9×10^9 $M^{-1}S^{-1}$) CA: PEG 0.1×10^9 $M^{-1}S^{-1}$).

The lowest quenching rate constant value of pyrene dissolved in CA: PEG is a direct consequence of the highest η of this system among the three (a similar trend follows at all investigated temperatures). The comparatively lower dynamic η of LA: PEG and HCA: PEG supports faster diffusion and dynamics of pyrene-nitromethane fluorophore-quencher pair, thereby increasing the rate of bimolecular quenching process in these DESs as compared to CA: PEG and other DESs of comparable η. Thus, the estimated k_q values follow the opposite trend as the η of DESs. These observations simply establish the role of dynamic η in facilitating the diffusion of the excited pyrene molecule and the nitromethane.

Further, the temperature-dependent behaviour of the bimolecular quenching rate constant for all three investigated DESs was studied to obtain the activation energy (E_a) for the quenching process. In each of the three DESs, the variation of ln k_q with 1/T adheres to the basic empirical Arrhenius equation, as shown in Fig. S3. It can be seen that E_a for CA: PEG is significantly higher in comparison to other documented systems [49]. This observation further supports the slow diffusion of nitromethane and excited pyrene in CA: PEG DES. The lower E_a values in LA: PEG and HCA: PEG imply more facile quenching dynamics in these DESs. Hence, the highest value of activation energy for CA: PEG is owed to its highest η. The E_a for LA: PEG and HCA: PEG is significantly lower than that of CA: PEG and hydrophilic reline (1:2) DESs (represented in Table S3 and Fig. S4). Notably, the E_a for LA: PEG and HCA: PEG are only moderately higher than Thy: DA and TBAC: DA hydrophobic DESs, attributed to the slight difference in the dynamic viscosities. These outcomes reveal that the diffusion process is rather facile within the HCA and LA-*based* DESs than the highly viscous CA-*based* DES. The recovered decay parameters τ_1, τ_2, α_1, and α_2, along with the goodness of the fit for all three investigated at different concentrations of quenchers [52], are shown in Tables S4–S6.

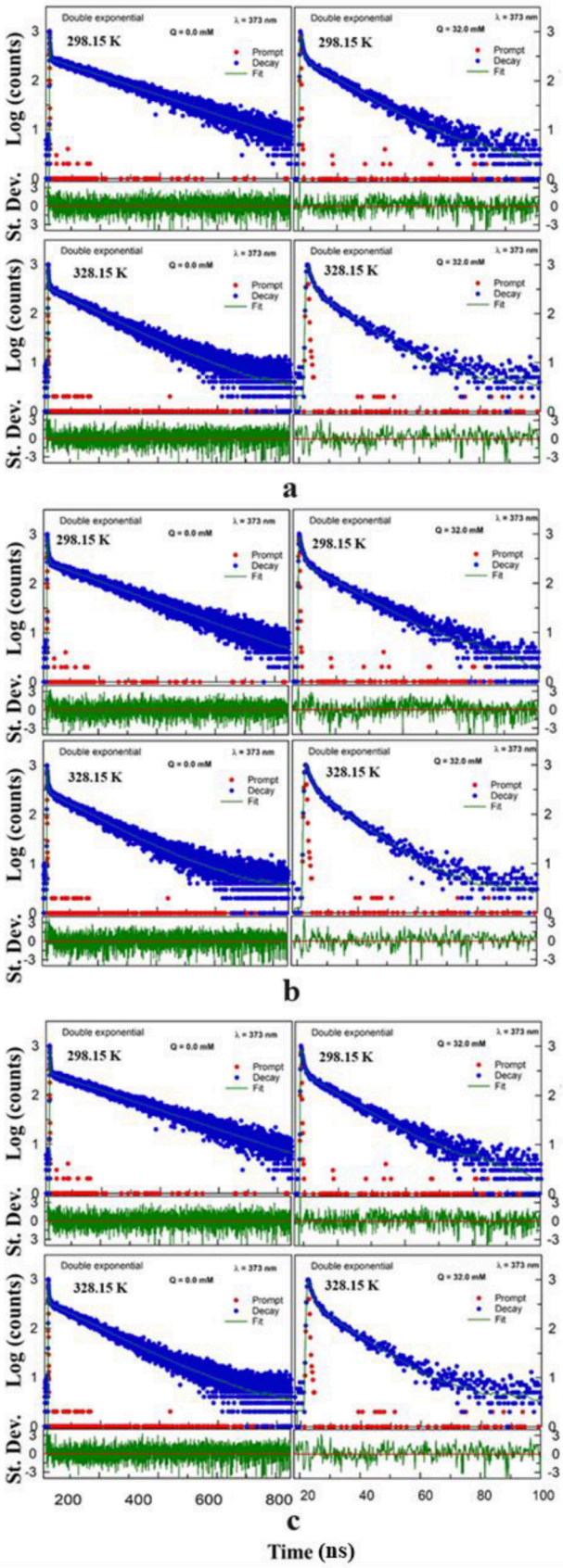


Fig. 8. Representative plot of the double-exponential decay fit (with 0 mM and 32.0 mM nitromethane) of the excited-state intensity decay of pyrene (10 μM; excitation with 340 nm nano-LED) in (a) LA: PEG; (b) HCA: PEG; and (c) CA: PEG at 298.15 and 328.15 K respectively.

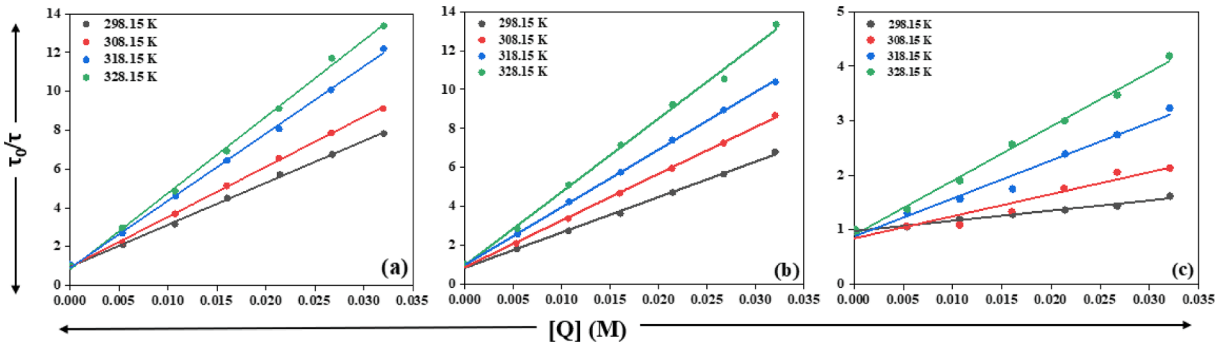


Fig. 9. Variation of τ0/τ versus [Q] for pyrene (10 μM) dissolved in (a) LA: PEG; (b) HCA: PEG; (c) CA: PEG. Solid lines represent the best fit to the Stern–Volmer equation.

Table 5
Estimated Stern–Volmer dynamic quenching rate constant (K_{SV}) and bimolecular quenching rate constant (k_q) for pyrene-nitromethane set in the PEGylated DESs at different temperatures.

T (K)	R^2	$K_{SV}(M^{-1})$	τ_0(ns)	$k_q \times 10^9$ ($M^{-1}s^{-1}$)
LA: PEG				
298.15	0.9990	216	213	1.0
308.15	0.9990	256	175	1.4
318.15	0.9983	339	156	2.2
328.15	0.9971	382	141	2.7
HCA: PEG				
298.15	0.9963	176	195	0.9
308.15	0.9981	236	180	1.3
318.15	0.9996	296	150	2.0
328.15	0.9964	378	136	2.8
338.15	0.9966	416	121	3.4
CA: PEG				
298.15	0.9908	18	250	0.1
308.15	0.8902	34	242	0.1
318.15	0.9670	64	237	0.3
328.15	0.9936	95	227	0.4
338.15	0.9935	179	219	0.8

4. Conclusion

The PEGylated DESs were effectively prepared using varied molar ratios. Studies on their thermophysical properties revealed that by modifying the type of HBDs and adjusting the molar ratios, the properties of the DESs can be tailored. The interaction between the HBAs and HBD is inferred from the FT-IR, ^{1}H- and ^{13}C NMR spectroscopy. The TGA technique is utilized to evaluate the thermal stability of prepared DESs. At the same time, DSC revealed the formation of the stable DES across a broad temperature range. For these DESs, the ρ and η decrease with increases the temperature. Also, the calculated thermophysical and its associated properties from the data of ρ and η. Here, the LA: PEG system exhibits high values of S° and U_{POT}, indicating that it is a highly stable DES compared to other DESs. We have employed Kamlet-Taft parameters and dynamic quenching of fluorescent probes to unveil some of the solvation characteristics of the DESs under investigation. Empirical dipolarity values (π*), supported by I_1/I_3 data of pyrene dissolved in all three DESs, were found to be highest for CA: PEG and followed by LA: PEG and HCA: PEG, respectively. The capacity of DESs investigated to donate H-bond to the solute is higher than some DESs and alcohols but slightly lower than water. Within PEGylated DESs investigated in this work, pyrene fluorescence and its quenching by nitromethane are helpful in revealing structural information and solute diffusion kinetics. Since these DESs do not support the excimer formation of pyrene, solvent–solute interactions can be studied through diffusion-controlled quenching of pyrene. Excited-state intensity decay of pyrene in the presence and the absence of a quencher shows a double exponential decay quencher regardless of temperature (298.15 K–338.15 K) and chemical constituents of DESs. The linear plots of τ_0/τ versus [Q] indicate that the fluorescence quenching is purely dynamic. As a result, the recovered Stern–Volmer dynamic quenching rate constant (K_{SV}) and bimolecular quenching rate constant (k_q) for the pyrene-nitromethane fluorophore-quencher pair in the temperature range of 298.15–338.15 K is lowest for CA: PEG owing to its highest η in this temperature range. Also, the higher η of CA: PEG demands higher activation energy E_a (50.0 ± 1.7 kJ mol^{-1}), even higher than Reline (42.4 ± 1.8 kJ mol^{-1}). In addition to providing insights into the structural characteristics and diffusion dynamics inside the hydrophobic DES, multidimensional pyrene fluorescence also establishes the DES as a solubilizing milieu, a cost-effective, ecologically friendly nonpolar media that can be used globally.

CRediT authorship contribution statement

Yagnik Vora: Validation, Resources, Methodology, Investigation, Formal analysis, Data curation. **Mithilesh Prajapati:** Validation, Investigation, Formal analysis, Data curation. **Vaishali Khokhar:** Validation, Investigation, Formal analysis, Data curation. **Sangesh P. Zodape:** Validation, Supervision, Methodology, Formal analysis, Data curation. **Kishant Kumar:** Writing – review & editing, Writing – original draft, Validation, Supervision. **Siddharth Pandey:** Writing – review & editing, Validation, Supervision, Methodology. **Ketan Kuperkar:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2025.127052.

Data availability

Data will be made available on request.

References

[1] A.P. Abbott, G. Capper, D.L. Davies, R.K. Rasheed, V. Tambyrajah, Novel solvent properties of choline chloride/urea mixtures, Chem. Commun. 1 (2003) 70–71, https://doi.org/10.1039/B210714G.

[2] S. Mor, R. Yadav, K. Bhakuni, P. Rawat, M. Bisht, N. Deenadayalu, P. Venkatesu, Unraveling the role of deep eutectic solvents with varying hydrogen-bond acceptors on the thermoresponsive polymer poly (N-isopropylacrylamide), J. Phys. Chem. B 128 (18) (2024) 4554–4565, https://doi.org/10.1021/acs.jpcb.4c00888.

[3] B.B. Hansen, S. Spittle, B. Chen, D. Poe, Y. Zhang, J.M. Klein, A. Horton, L. Adhikari, T. Zelovich, B.W. Doherty, B. Gurkan, E.J. Maginn, A. Ragauskas, M. Dadmun, T.A. Zawodzinski, G.A. Baker, M.E. Tuckerman, R.F. Savinell, J. R. Sangoro, Deep eutectic solvents: a review of fundamentals and applications, Chem. Rev. 121 (3) (2021) 1232–1285, https://doi.org/10.1021/acs.chemrev.0c00385.

[4] O.V. Morozova, I.S. Vasil'eva, G.P. Shumakovich, E.A. Zaitseva, A.I. Yaropolov, Deep Eutectic solvents for biotechnology applications, Biochemistry (Mosc) 88 (Suppl. 1) (2023) S150–S175, https://doi.org/10.1134/S0006297923140092.

[5] Y. Wang, N. Sun, H. Cheng, S. Zhou, X. Ouyang, X. Zhang, N. Ma, Highly sensitive flexible thermal sensors based on a kind of MXene/DES inks, ACS Appl. Energy Mater. 5 (2) (2023) 1252–1261, https://doi.org/10.1021/acsaelm.2c01710.

[6] Y. Cao, Y. Li, M. Sun, Y. Xu, L. Chen, Unexpectedly superhigh toxicity of superbase-derived deep eutectic solvents albeit high efficiency for CO_2 capture and conversion, Ind. Eng. Chem. Res. 62 (7) (2023) 3338–3347, https://doi.org/10.1021/acs.iecr.2c04564.

[7] J.B. Marlow, R. Atkin, G.G. Warr, How does nanostructure in ionic liquids and hybrid solvents affect surfactant self-assembly? J. Phys. Chem. B 127 (7) (2023) 1490–1498, https://doi.org/10.1021/acs.jpcb.2c07458.

[8] G.A.L. e Souza, M. Di Pietro, V. Vanoli, W. Panzeri, F. Briatico-Vangosa, F. Castiglione, A. Mele, Hydrophobic eutectogels: a new outfit for non-ionic eutectic solvents, Mater. Today Chem. 29 (2023) 101402–101408, https://doi.org/10.1016/j.mtchem.2023.101402.

[9] N. Akbar, N.A. Khan, T. Ibrahim, M. Khamis, A.S. Khan, A.M. Alharbi, H. Alfahemi, R. Siddiqui, Antimicrobial activity of novel deep eutectic solvents, Sci. Pharm. 91 (1) (2023) 9–23, https://doi.org/10.3390/scipharm91010009.

[10] H. Shaaban, Sustainable dispersive liquid–liquid microextraction method utilizing a natural deep eutectic solvent for determination of chloramphenicol in honey: assessment of the environmental impact of the developed method, RSC Adv. 13 (2023) 5058–5069, https://doi.org/10.1039/D2RA08221G.

[11] P.J. Smith, J.C. Goeltz, Proton-coupled electron transfer and substituent effects in catechol-based deep eutectic solvents: gross and fine tuning of redox activity, J. Phys. Chem. B 121 (48) (2017) 10974–10978, https://doi.org/10.1021/acs.jpcb.7b10169.

[12] D.O. Abranches, J.A.P. Coutinho, Type V deep eutectic solvents: design and applications, Curr. Opin. Green Sustain. Chem. 35 (2022) 100612–100619, https://doi.org/10.1016/j.cogsc.2022.100612.

[13] A. Singh, R. Walvekar, M. Khalid, W.Y. Wong, T.C.S.M. Gupta, Thermophysical properties of glycerol and polyethylene glycol (PEG 600) based DES, J. Mol. Liq. 252 (2018) 439–444, https://doi.org/10.1016/j.molliq.2017.10.030.

[14] D.O. Abranches, M.A. Martins, L.P. Silva, N. Schaeffer, S.P. Pinho, J.A.P. Coutinho, Phenolic hydrogen bond donors in the formation of non-ionic deep eutectic solvents: the quest for type V DES, Chem. Commun. 55 (69) (2019) 10253–10256, https://doi.org/10.1039/C9CC04846D.

[15] F. Lima, J. Gouvenaux, L.C. Branco, A.J. Silvestre, I.M. Marrucho, Towards a sulfur clean fuel: deep extraction of thiophene and dibenzothiophene using polyethylene glycol-based deep eutectic solvents, Fuel 234 (2018) 414–421, https://doi.org/10.1016/j.fuel.2018.07.043.

[16] Y. Chen, C. Liu, Y. Wang, Y. Tian, Y. Li, M. Feng, Y. Guo, J. Han, T. Mu, Efficient recovery of valuable metals from lithium-ion battery cathodes using phytic acid-based deep eutectic solvents at a mild temperature, Energy Fuel 37 (7) (2023) 5361–5369, https://doi.org/10.1021/acs.energyfuels.3c00313.

[17] Y. Chen, Q. Wang, Z. Liu, Z. Li, W. Chen, L. Zhou, J. Qin, Y. Meng, T. Mu, Vaporization enthalpy, long-term evaporation and evaporation mechanism of polyethylene glycol-based deep eutectic solvents, New J. Chem. 44 (22) (2020) 9493–9501, https://doi.org/10.1039/D0NJ01601B.

[18] J. Jiang, C. Yan, X. Zhao, H. Luo, Z. Xue, T. Mu, A PEGylated deep eutectic solvent for controllable solvothermal synthesis of porous $NiCo_2S_4$ for efficient oxygen evolution reaction, Green Chem. 19 (13) (2017) 3023–3031, https://doi.org/10.1039/C7GC01012E.

[19] J.R. Lakowicz, Principles of Fluorescence Spectroscopy, Springer, 2006 doi: 10.1007/978-0-387-46312-4_2.

[20] B. Valeur, M.N. Berberan-Santos, Molecular Fluorescence: Principles and Applications, John Wiley & Sons, 2012 doi: 10.1002/9783527650002.

[21] C. Florindo, F.S. Oliveira, L.P.N. Rebelo, A.M. Fernandes, I.M. Marrucho, Insights into the synthesis and properties of deep eutectic solvents based on cholinium chloride and carboxylic acids, ACS Sustain. Chem. Eng. 2 (10) (2014) 2416–2425, https://doi.org/10.1021/sc500439w.

[22] C.V. Rajput, R.B. Mukherjee, N.V. Sastry, N.P. Chikhaliya, Epoxidized cassia fistula seed oil as bio-based plasticizer for poly (vinyl chloride) soft films, ACS Appl. Polym. Mater. 4 (12) (2022) 8926–8941, https://doi.org/10.1021/acsapm.2c01333.

[23] A.K. Padhan, D. Sharma, T.S. Thomas, A.P. Sinha, A.N. Mallick, D. Mandal, Rapid self-healing and superior toughness in ionically crosslinked polymer ionogels and strain sensing applications, J. Mater. Chem. A 12 (16) (2024) 9508–9517, https://doi.org/10.1039/D3TA07277K.

[24] N.S. Vrandečić, M. Erceg, M. Jakić, I. Klarić, Kinetic analysis of thermal degradation of poly(ethylene glycol) and poly(ethylene oxide)s of different molecular weight, Thermochim. Acta 498 (1–2) (2010) 71–80, https://doi.org/10.1016/j.tca.2009.10.005.

[25] X. Zhao, X. Lan, D. Yu, H. Fu, Z. Liu, T. Mu, Deep eutectic-solvothermal synthesis of nanostructured Fe_3S_4 for electrochemical N_2 fixation under ambient conditions, Chem. Commun. 54 (92) (2018) 13010–13013, https://doi.org/10.1039/C8CC08045C.

[26] W. Chen, X. Bai, Z. Xue, H. Mou, J. Chen, Z. Liu, T. Mu, The formation and physicochemical properties of PEGylated deep eutectic solvents, New J. Chem. 43 (22) (2019) 8804–8810, https://doi.org/10.1039/C9NJ02196E.

[27] H. Ghaedi, M. Ayoub, S. Sufian, B. Lal, Y. Uemura, Thermal stability and FT-IR analysis of phosphonium-based deep eutectic solvents with different hydrogen bond donors, J. Mol. Liq. 242 (2017) 395–403, https://doi.org/10.1016/j.molliq.2017.07.016.

[28] N. Delgado-Mellado, M. Larriba, P. Navarro, V. Rigual, M. Ayuso, J. García, F. Rodríguez, Thermal stability of choline chloride deep eutectic solvents by TGA/FTIR-ATR analysis, J. Mol. Liq. 260 (2018) 37–43, https://doi.org/10.1016/j.molliq.2018.03.076.

[29] D.J.G.P. van Osch, C.H.J.T. Dietz, J. van Spronsen, M.C. Kroon, F. Gallucci, M. van Sint Annaland, R. Tuinier, A search for natural hydrophobic deep eutectic solvents based on natural components, ACS Sustain. Chem. Eng. 7 (3) (2019) 2933–2942, https://doi.org/10.1021/acssuschemeng.8b03520.

[30] H. Cruz, N. Jordao, A.L. Pinto, M. Dionisio, L.A. Neves, L.C. Branco, Alkaline iodide-based deep eutectic solvents for electrochemical applications, ACS Sustain. Chem. Eng. 8 (29) (2020) 10653–10663, https://doi.org/10.1021/acssuschemeng.9b06733.

[31] C. Florindo, M.M. Oliveira, L.C. Branco, I.M. Marrucho, Carbohydrates-based deep eutectic solvents: thermophysical properties and rice straw dissolution, J. Mol. Liq. 247 (2017) 441–447, https://doi.org/10.1016/j.molliq.2017.09.026.

[32] A.K. Jangir, P. Sethy, G. Verma, P. Bahadur, K. Kuperkar, An inclusive thermophysical and rheology portrayal of deep eutectic solvents (DES) for metal oxides dissolution enhancement, J. Mol. Liq. 332 (2021) 115909–115920, https://doi.org/10.1016/j.molliq.2021.115909.

[33] J. Gao, X. Yang, Z. Xing, X. Song, Y. Liu, Z. Wang, G. Deng, X. Zhao, Physicochemical and thermodynamic properties of binary amine-based deep eutectic solvents for carbon capture, J. Mol. Liq. 399 (2024) 124346–124360, https://doi.org/10.1016/j.molliq.2024.124346.

[34] L. Glasser, Lattice and phase transition thermodynamics of ionic liquids, Thermochim. Acta 421 (1–2) (2004) 87–93, https://doi.org/10.1016/j.tca.2004.03.015.

[35] D.W. Fang, W. Guan, J. Tong, Z.W. Wang, J.Z. Yang, Study on physicochemical properties of ionic liquids based on alanine [$C_n$mim][Ala](n= 2, 3, 4, 5, 6), J. Phys. Chem. B 112 (25) (2008) 7499–7505, https://doi.org/10.1021/jp801269u.

[36] H.H. Al-Kayiem, T.L. Oladosu, S.I. Gilani, A.T. Baheta, Molecular dynamics of binary deep eutectic solvents as biocompatible working fluids in heat and mass transfer systems, J. Mol. Liq. 342 (2021) 117493–117507, https://doi.org/10.1016/j.molliq.2021.117493.

[37] M.K. AlOmar, M. Hayyan, M.A. Alsaadi, S. Akib, A. Hayyan, M.A. Hashim, Glycerol-based deep eutectic solvents: physical properties, J. Mol. Liq. 215 (2016) 98–103, https://doi.org/10.1016/j.molliq.2015.11.032.

[38] A.K. Dwamena, D.E. Raynie, Solvatochromic parameters of deep eutectic solvents: effect of different carboxylic acids as hydrogen bond donor, J. Chem. Eng. Data 65 (2) (2020) 640–646, https://doi.org/10.1021/acs.jced.9b00872.

[39] R.W. Taft, M.J. Kamlet, The solvatochromic comparison method. 2. The. alpha.-scale of solvent hydrogen-bond donor (HBD) acidities, J. Am. Chem. Soc. 98 (10) (1976) 2886–2894.

[40] M.J. Kamlet, R.W. Taft, The solvatochromic comparison method. I. The. beta.-scale of solvent hydrogen-bond acceptor (HBA) basicities, J. Am. Chem. Soc. 98 (2) (1976) 377–383.

[41] J.M. Lee, S. Ruckes, J.M. Prausnitz, Solvent polarities and Kamlet–Taft parameters for ionic liquids containing a pyridinium cation, J. Phys. Chem. B 112 (5) (2008) 1473–1476, https://doi.org/10.1021/jp076895k.

[42] M. Aryafard, M. Abbasi, D. Řeha, A.R. Harifi-Mood, B. Minofar, Experimental and theoretical investigation of solvatochromic properties and ion solvation structure in DESs of reline, glyceline, ethaline and their mixtures with PEG 400, J. Mol. Liq. 284 (2019) 59–67, https://doi.org/10.1016/j.molliq.2019.03.149.

[43] M. Aryafard, A. Karimi, A.R. Harifi-Mood, B. Minofar, Molecular dynamics simulations, solvatochromic parameters, and preferential solvation in aqueous solutions of ethaline, ethylene glycol, and choline chloride, J. Chem. Eng. Data 65 (9) (2020) 4556–4566, https://doi.org/10.1021/acs.jced.0c00381.

[44] P.G. Jessop, D.A. Jessop, D. Fu, L. Phan, Solvatochromic parameters for solvents of interest in green chemistry, Green Chem. 14 (5) (2012) 1245–1259, https://doi.org/10.1039/C2GC16670D.

[45] A.R.R. Teles, E.V. Capela, R.S. Carmo, J.A. Coutinho, A.J. Silvestre, M.G. Freire, Solvatochromic parameters of deep eutectic solvents formed by ammonium-based salts and carboxylic acids, Fluid Ph. Equilib. 448 (2017) 15–21, https://doi.org/10.1016/j.fluid.2017.04.020.

[46] A. Kadyan, S. Pandey, Florescence quenching within lithium salt-added ionic liquid, J. Phys. Chem. B 122 (19) (2018) 5106–5113, https://doi.org/10.1021/acs.jpcb.8b02723.
[47] A. Kadyan, S. Pandey, Lithium bis (trifluoromethylsulfonyl) imide-added ionic liquid 1-ethyl-3-methylimidazolium bis (trifluoromethylsulfonyl) imide mixture: densities and dynamic viscosities in the temperature range (298.15–358.15) K, J. Chem. Thermodyn. 116 (2018) 159–165, https://doi.org/10.1016/j.jct.2017.08.033.
[48] A. Patra, V. Khokhar, S. Pandey, Decanoic acid based deep eutectic solvents: constituent-controlled physicochemical properties and solvatochromic probe behavior, Ind. Eng. Chem. Res. 61 (39) (2022) 14702–14710, https://doi.org/10.1021/acs.iecr.2c02608.
[49] D. Dhingra, Bhawna, A. Pandey, S. Pandey, Pyrene fluorescence to probe a lithium chloride-added (choline chloride + urea) deep eutectic solvent, J. Phys. Chem. B 123 (14) (2019) 3103–3111, https://doi.org/10.1021/acs.jpcb.9b01193.
[50] S. Juneja, S. Pandey, Fluorescence of pyrene and its derivatives to reveal constituent and composition dependent solvation within hydrophobic deep eutectic solvents, PCCP 25 (17) (2023) 11998–12012, https://doi.org/10.1039/D3CP01207G.
[51] D. Dhingra, Bhawna, A. Pandey, S. Pandey, Fluorescence quenching by nitro compounds within a hydrophobic deep eutectic solvent, J. Phys. Chem. B 124 (20) (2020) 4164–4173, https://doi.org/10.1021/acs.jpcb.0c02231.
[52] D. Dhingra, S. Pandey, Effect of lithium chloride on the density and dynamic viscosity of choline chloride/urea deep eutectic solvent in the temperature range (303.15–358.15) K, J. Chem. Thermodyn. 130 (2019) 166–172, https://doi.org/10.1016/j.jct.2018.10.003.